Problem Set for 07 April 2025

Exercise Let V be a vector space over $\mathbb{F}$, and define $\operatorname{End}(V) := \operatorname{Hom}_{\mathbb{F}}(V, V)$. Define:

- $f \in \operatorname{End}(V)$ as *almost zero* (denoted $f \in \operatorname{AZ}(V)$), provided $\dim \operatorname{im}(f) < \infty$.
- $f \in \operatorname{End}(V)$ as *almost an isomorphism* (denoted $f \in \operatorname{AI}(V)$), provided $\dim \ker f + \dim \frac{V}{\operatorname{im}(f)} < \infty$.

Now finish the following.

1. If $\dim V < \infty$, determine $\operatorname{AZ}(V)$ and $\operatorname{AI}(V)$.

> $\operatorname{AZ}(V) = \operatorname{AI}(V) = \operatorname{End}(V)$.

2. If $V = \mathbb{R}[x]$, find an injective $f \in \operatorname{AI}(V)$ and a surjective $g \in \operatorname{AI}(V)$. Additionally, find some $h \notin \operatorname{AI}(V) \cup \operatorname{AZ}(V)$.

> *f 右移, g 左移. 将 $\mathbb{R}[x]$ 依照奇偶函数写作 $V_1 \oplus V_2$, 则*
>
> $$\varphi : V_1 \oplus V_2 \to V_1 \oplus V_2, \quad (f, g) \mapsto (f, 0)$$
>
> *满足 $\dim \ker \varphi, \dim \operatorname{im} \varphi, \dim \frac{\mathbb{R}[x]}{\operatorname{im} \varphi} = \infty$.*

3. Determine whether $\operatorname{AZ}(V)$ is a subspace of $\operatorname{End}(V)$. What about $\operatorname{AI}(V)$?

> *先看 $\operatorname{AZ}(V)$, 下分别证明 $\operatorname{AZ}(V)$ 对数乘和加法封闭.*
>
> 1. *$f \in \operatorname{AZ}(V)$, 则 $\operatorname{im}(\lambda f) \subset \operatorname{im}(f)$ 是有限维的.*
> 2. *$f, g \in \operatorname{AZ}(V)$, 则 $\operatorname{im}(f+g) \subset \operatorname{im}(f) + \operatorname{im}(g)$ 是有限维的.*
>
> *$0 \notin \operatorname{AI}(V)$, 从而 $\operatorname{AI}(V)$ 不是线性子空间.*

4. Show that for arbitrary $f, g \in \operatorname{AZ}(V)$, one has $g \circ f \in \operatorname{AZ}(V)$.

> *下一问的弱化.*

5. Prove that $\operatorname{AZ}(V)$ is an **ideal** in $\operatorname{End}(V)$. The term **ideal** is defined in *20 March 2025*.

> *若 $f \in \operatorname{AZ}(V)$, 由满射*
>
> $$\varphi : \operatorname{im}(f) \twoheadrightarrow \operatorname{im}(g \circ f), \quad f(v) \mapsto g(f(v)),$$

此时 $\dim\operatorname{im}(g\circ f)\le\operatorname{im}(f)$. 从而 $g\circ f\in\mathrm{AZ}(V)$.
若 $g\in\mathrm{AZ}(V)$, 显然 $\operatorname{im}(g\circ f)\subset\operatorname{im}(g)$ 是有限维的. 从而 $g\circ f\in\mathrm{AZ}(V)$.
$\mathrm{AZ}(V)$ 是线性空间, 且

$$\mathrm{AZ}(V)\circ\mathrm{End}(V)\subset\mathrm{AZ}(V)\supset\mathrm{End}(V)\circ\mathrm{AZ}(V).$$

从而是理想.

6. Show that for arbitrary $f,g\in\mathrm{AI}(V)$, one has $g\circ f\in\mathrm{AI}(V)$.

只需证明以下事实.

1. $\dim\ker f+\dim\ker g\ge\dim\ker(g\circ f)$.
2. $\dim\frac{V}{\operatorname{im}(f)}+\dim\frac{V}{\operatorname{im}(g)}\ge\dim\frac{V}{\operatorname{im}(g\circ f)}$.

第一问, 构造

$$f_s:\ker(g\circ f)\to\ker(g),\quad x\mapsto f(x).$$

这一映射的 image 是 $\ker g$ 的子空间, kernel 是 $\ker(g\circ f)\cap\ker f=\ker f$. 从而

$$\dim\ker(g\circ f)=\dim\ker f+\dim\ker g-\dim\frac{\ker g}{f(\ker(g\circ f))}.$$

由 $\dim\frac{\ker g}{f(\ker(g\circ f))}\le\dim\ker g<\infty$, 左式是有限的.
第二问, 构造

$$g_q:\frac{V}{\operatorname{im}(f)}\to\frac{V}{\operatorname{im}(g\circ f)},\quad (v+f(V))\mapsto(g(v)+g(f(V))).$$

这一 g_q 是良定义的, 因为

$$\Phi:V\to\frac{V}{\operatorname{im}(g\circ f)},\quad v\mapsto(g(v)+g(V))$$

的 kernel 包含了 $\operatorname{im}(f)$, 从而 Φ 可以定义在商空间上, 得 g_q. 特别地,

> $$\begin{aligned}\frac{V/\mathrm{im}(g\circ f)}{\mathrm{im}(g_q)} &= \frac{V/\mathrm{im}(g\circ f)}{\mathrm{im}(\Phi)} && \text{由 } \Phi = g_q \circ \text{“某满射”} \\ &= \frac{V/\mathrm{im}(g\circ f)}{\{g(v)+\mathrm{im}(g\circ f)\}} && \text{由定义} \\ &= \frac{V/\mathrm{im}(g\circ f)}{\mathrm{im}(g)/\mathrm{im}(g\circ f)} && \text{写作商空间的子空间} \\ &= \frac{V}{\mathrm{im}(g)} && \text{同构定理.}\end{aligned}$$
>
> *因此* $g_q(\frac{V}{\mathrm{im}(f)})$ *是满射* $\frac{V}{\mathrm{im}(g\circ f)} \twoheadrightarrow \frac{V}{\mathrm{im}(g)}$ *的 kernel. 从而*
>
> $$\dim\frac{V}{\mathrm{im}(g\circ f)} = \dim\frac{V}{\mathrm{im}(g)} + \dim\frac{V}{\mathrm{im}(f)} - \dim\ker g_q.$$
>
> *由* $\dim\ker g_q \leq \dim\frac{V}{\mathrm{im}(f)} < \infty$, *左式是有限的.*

7. Demonstrate that if any two elements of $\{f, g, g\circ f\}$ belong to $\mathrm{AI}(V)$, then so does the third.

> *Hint: For* $\Phi(f) := \dim\ker f + \dim\frac{V}{\mathrm{im}(f)} < \infty$, *show that* $\Phi(f\circ g) = \Phi(f) + \Phi(g)$.
>
> *Hint 中的等式成立, 当且仅当上述*
>
> $$\dim\frac{\ker g}{f(\ker(g\circ f))} = \dim\ker g_q.$$
>
> *实际上,*
>
> 1. $\ker g_q = \frac{\ker g + \mathrm{im}\, f}{\mathrm{im}\, f}$;
> 2. $\frac{\ker g}{f(\ker(g\circ f))} = \frac{\ker g}{\mathrm{im}\, f\cap\ker g}$.
>
> *得证.*

8. Show that for arbitrary $f \in \mathrm{AZ}(V)$, one has $(\mathrm{id}_V + f) \in \mathrm{AI}(V)$.

> *若* $(\mathrm{id}_V + f)(v) = 0$, *则* $v \in \mathrm{im}\, f$. *从而* $\ker(\mathrm{id}_V + f) \subset \mathrm{im}\, f$ *是有限维的.*
>
> *对偶地, 只需证明存在满射* $\mathrm{im}\, f \twoheadrightarrow \frac{V}{\mathrm{im}(\mathrm{id}_V + f)}$, *依照同构定理, 即证* $\ker f \subset \mathrm{im}(\mathrm{id}_V + f)$. *若* $f(v) = 0$, *则* $v = (\mathrm{id}_V + f)(v)$, *得证.*

9. **(Optional)** By Axiom of Choice, every subspace has a complementary subspace. Prove that $f \in \mathrm{AI}(V)$ can always be expressed as the summation of a map of $\mathrm{AZ}(V)$-type and an automorphism (i.e., auto = iso + endo). Conversely, show that any such summation belongs to $\mathrm{AI}(V)$.

> *对 f, 考虑*
>
> $$V = \ker f \oplus V_1 \xrightarrow{f} \mathrm{im}(f) \oplus V_2 = V.$$
>
> *以上诱导的 $V_1 \to \mathrm{im}\, f$ 是同构. 由于 $\ker f$ 和 V_2 是有限维的, 任何通过 $\ker f$ 或 V_2 分解的映射都是 $\mathrm{AZ}(V)$-类型的. 可以将 f 写作列向量之间的矩阵, 即*
>
> $$\begin{pmatrix} a & b \\ c & d \end{pmatrix} : \ker f \oplus V_1 \to \mathrm{im}(f) \oplus V_2, \quad (x, y) \mapsto (a(x) + b(y), c(x) + d(y)).$$
>
> *此时* $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} a & 0 \\ c & d \end{pmatrix}$ *就是所求的分解.*

10. **(Optional)** Show that $\frac{\mathrm{End}(V)}{\mathrm{AZ}(V)}$ is a quotient ring with a well-defined subset $\frac{\mathrm{AI}(V)}{\mathrm{AZ}(V)}$, consisting precisely of the invertible elements. In this quotient ring, the operators $\frac{\mathrm{d}}{\mathrm{d}\,x}(-)$ and $\int_0^x (-)\,\mathrm{d}\,x$ are mutually inverse.

> *本小问面向熟悉环论的同学, 其实没什么新东西.*

Problem (Optional) Let V be a vector space over $\mathbb{F}$. We define

$$\prod_{i\in I} V := V \underbrace{\times \cdots \times}_{I\text{-copies}} V \simeq \mathrm{Hom}_{\mathrm{Sets}}(I, V)$$

as an $\mathbb{F}$-vector space whose elements are sequences $(v_i)_{i\in I}$. Informally, $\prod_{i\in I} V$ represents the space of sequences indexed by I with values in V.

1. Define $e_i := (\ldots, 0, 0, \underbrace{1_{\mathbb{F}}}_{i\text{-th entry}}, 0, 0, \ldots) \in \prod_{i\in I} \mathbb{F}$. Determine when $\{e_i\}_{i\in I}$ forms a basis of $\prod_{i\in I} \mathbb{F}$.

> *当且仅当 I 有限.*

2. Describe $\mathrm{span}(\{e_i\}_{i\in I})$ in natural language.

> $\prod_{i\in I} \mathbb{F}$ *中有且仅有有限项非零的东西.*

3. Substituting I with $V_{\text{as a set}}$, the underlying set of V as an $\mathbb{F}$-vector space, prove that π is a linear surjection and determine the equivalence relation $R_{\mathbb{F}}$ as defined in 24 *March* 2025:

$$\pi : \mathrm{span}(\{e_v\}_{v\in V_{\text{as a set}}}) \to V, \quad e_v \mapsto v.$$

> *映射层面,* $\mathrm{id}_{\mathrm{V}_{\text{as a set}}} : \{e_v\}_{v\in V_{\text{as a set}}} \to V$ *是恒等的. 由于* $\{e_v\}_{v\in V_{\text{as a set}}}$ *是* $\prod_V \mathbb{F}$ *中的线性无关组, 自然存在唯一定义的*
>
> $$\pi : \mathrm{span}(\{e_v\}_{v\in V_{\text{as a set}}}) \to V, \quad e_v \mapsto v.$$
>
> *使得限制在基上的映射为* $\pi|_{\{e_v\}_{v\in V_{\text{as a set}}}} = \mathrm{id}_{\mathrm{V}_{\text{as a set}}}$.
>
> *这时候,* $R_{\mathbb{F}}$ *就是* V *中本有的线性组合关系.*

Remark 以上问题就是将线性空间的定义重新表述了一遍: 结构 = 自由定义的结构商去关系.